



CAT 2006 Question Paper

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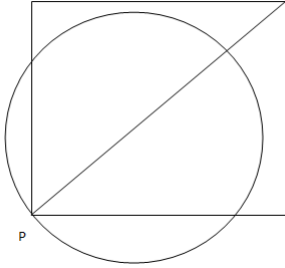
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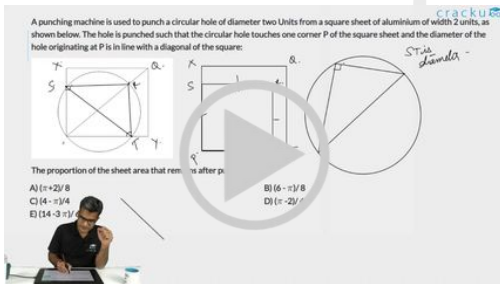
Instructions [1 - 2]

A punching machine is used to punch a circular hole of diameter two Units from a square sheet of aluminium of width 2 units, as shown below. The hole is punched such that the circular hole touches one corner P of the square sheet and the diameter of the hole originating at P is in line with a diagonal of the square:



1. The proportion of the sheet area that remains after punching is:

- A $(\pi+2)/8$
- B $(6-\pi)/8$
- C $(4-\pi)/4$
- D $(\pi-2)/4$
- E $(14-3\pi)/6$



VIDEO SOLUTION



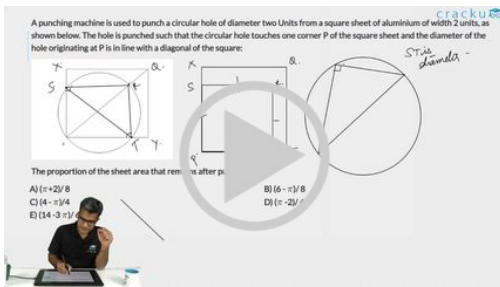
CAT Mock Test

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2. Find the area of the part of the circle (round punch) falling outside the square sheet.

- A $\pi/4$
- B $(\pi-1)/2$
- C $(\pi-1)/4$
- D $(\pi-2)/2$
- E $(\pi-2)/4$



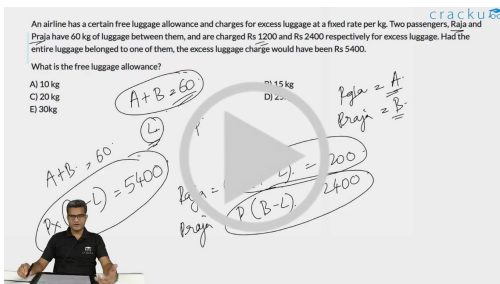
VIDEO SOLUTION

Instructions [3 - 4]

An airline has a certain free luggage allowance and charges for excess luggage at a fixed rate per kg. Two passengers, Raja and Praja have 60 kg of luggage between them, and are charged Rs 1200 and Rs 2400 respectively for excess luggage. Had the entire luggage belonged to one of them, the excess luggage charge would have been Rs 5400.

3. What is the weight of Praja's luggage?

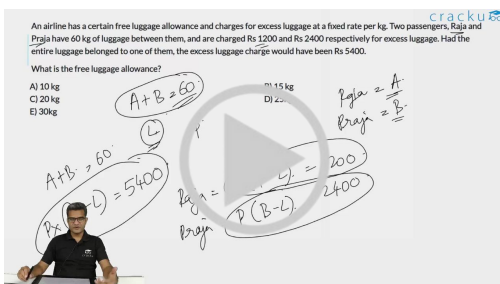
- A 20 kg
- B 25 kg
- C 30 kg
- D 35 kg
- E 40 kg



VIDEO SOLUTION

4. What is the free luggage allowance?

- A 10 kg
- B 15 kg
- C 20 kg
- D 25kg
- E 30kg



VIDEO SOLUTION



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5. If $x = -0.5$, then which of the following has the smallest value?

- A $2^{1/x}$
- B $1/x$
- C $1/x^2$
- D 2^x
- E $1/\sqrt{-x}$



VIDEO SOLUTION

6. Which among $2^{1/2}$, $3^{1/3}$, $4^{1/4}$, $6^{1/6}$, and $12^{1/12}$ is the largest?

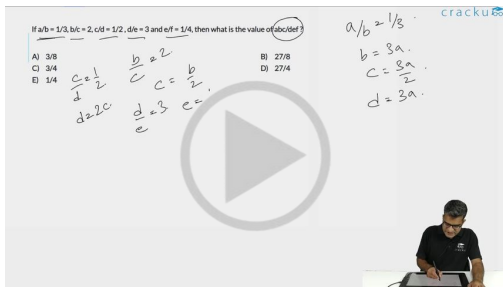
- A $2^{1/2}$
- B $3^{1/3}$
- C $4^{1/4}$
- D $6^{1/6}$
- E $12^{1/12}$



VIDEO SOLUTION

7. If $a/b = 1/3$, $b/c = 2$, $c/d = 1/2$, $d/e = 3$ and $e/f = 1/4$, then what is the value of abc/def ?

- A $3/8$
- B $27/8$
- C $3/4$
- D $27/4$
- E $1/4$



VIDEO SOLUTION

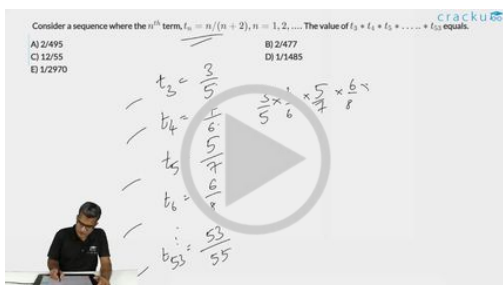


8. The length, breadth and height of a room are in the ratio 3:2:1. If the breadth and height are halved while the length is doubled, then the total area of the four walls of the room will

- A remain the same
- B decrease by 13.64%
- C decrease by 15%
- D decrease by 18.75%
- E decrease by 30%

9. Consider a sequence where the n^{th} term, $t_n = n/(n+2)$, $n = 1, 2, \dots$. The value of $t_3 * t_4 * t_5 * \dots * t_{53}$ equals.

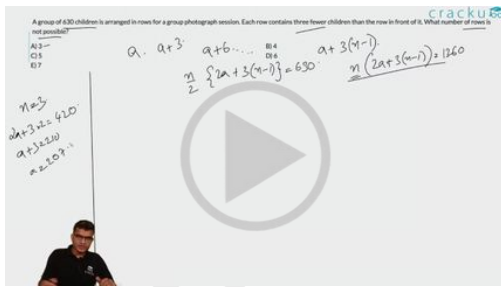
- A 2/495
- B 2/477
- C 12/55
- D 1/1485
- E 1/2970



VIDEO SOLUTION

10. A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. What number of rows is not possible?

- A 3
- B 4
- C 5
- D 6
- E 7



VIDEO SOLUTION



CAT Syllabus PDF



11. What are the values of x and y that satisfy both the equations?

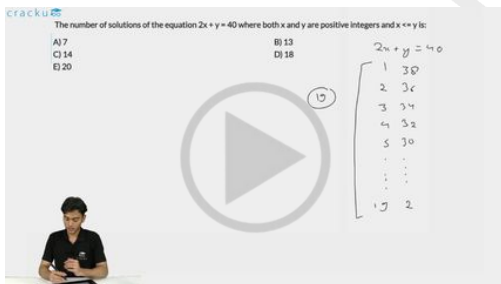
$$2^{0.7x} * 3^{-1.25y} = 8\sqrt{6}/27$$

$$4^{0.3x} * 9^{0.2y} = 8 * 81^{1/5}$$

- A $x = 2, y = 5$
- B $x = 2.5, y = 6$
- C $x = 3, y = 5$
- D $x = 3, y = 4$
- E $x = 5, y = 2$

12. The number of solutions of the equation $2x + y = 40$ where both x and y are positive integers and $x \leq y$ is:

- A 7
- B 13
- C 14
- D 18
- E 20



VIDEO SOLUTION

13. A survey was conducted of 100 people to find out whether they had read recent issues of Golmal, a monthly magazine. The summarized information regarding readership in 3 months is given below:

Only September: 18;

September but not August: 23;

September and July: 8;

September: 28;

July: 48;

July and August: 10;

none of the three months: 24

What is the number of surveyed people who have read exactly two consecutive issues (out of the three)?

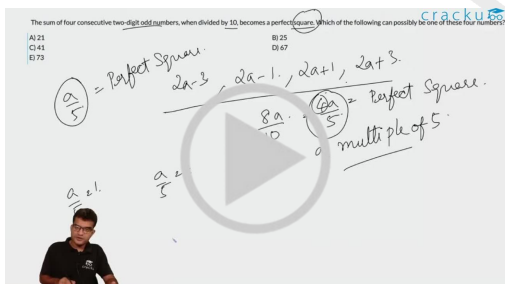
- A 7
- B 9

- C 12
D 14
E 17



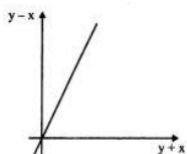
14. The sum of four consecutive two-digit odd numbers, when divided by 10, becomes a perfect square. Which of the following can possibly be one of these four numbers?

- A 21
B 25
C 41
D 67
E 73

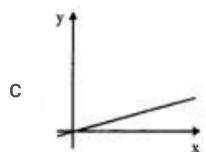
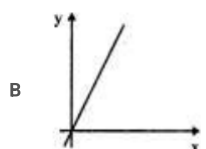
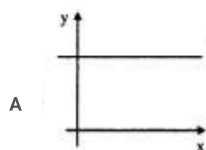


VIDEO SOLUTION

15. The graph of $y - x$ (on the y axis) against $y + x$ (on the x axis) is as shown below. (All graphs in this question are drawn to scale and the same scale and the same scale has been used on each axis.)



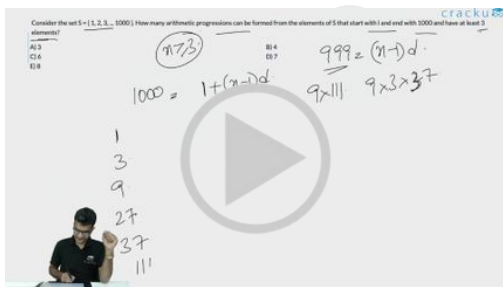
Which of the following shows the graph of y against x ?



16. Consider the set $S = \{1, 2, 3, \dots, 1000\}$. How many arithmetic progressions can be formed from the elements of S that start with 1 and end with 1000 and have at least 3 elements?

- A 3

- B 4
- C 6
- D 7
- E 8

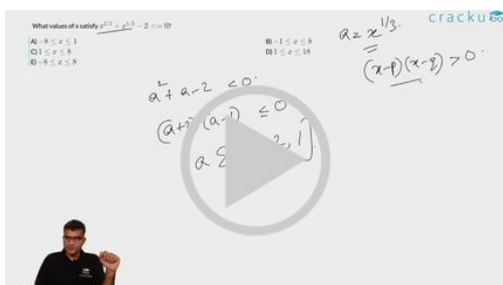


VIDEO SOLUTION



17. What values of x satisfy $x^{2/3} + x^{1/3} - 2 \leq 0$?

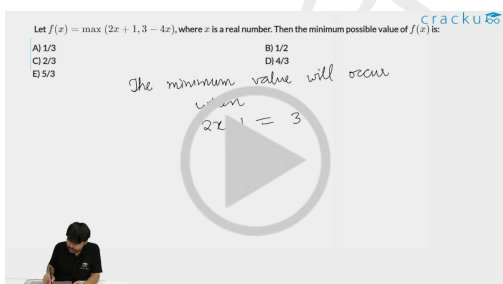
- A $-8 \leq x \leq 1$
- B $-1 \leq x \leq 8$
- C $1 \leq x \leq 8$
- D $1 \leq x \leq 18$
- E $-8 \leq x \leq 8$



VIDEO SOLUTION

18. Let $f(x) = \max(2x + 1, 3 - 4x)$, where x is a real number. Then the minimum possible value of $f(x)$ is:

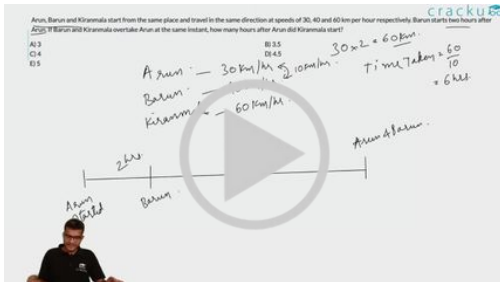
- A $1/3$
- B $1/2$
- C $2/3$
- D $4/3$
- E $5/3$



VIDEO SOLUTION

19. Arun, Barun and Kiranmala start from the same place and travel in the same direction at speeds of 30, 40 and 60 km per hour respectively. Barun starts two hours after Arun. If Barun and Kiranmala overtake Arun at the same instant, how many hours after Arun did Kiranmala start?

- A 3
- B 3.5
- C 4
- D 4.5
- E 5



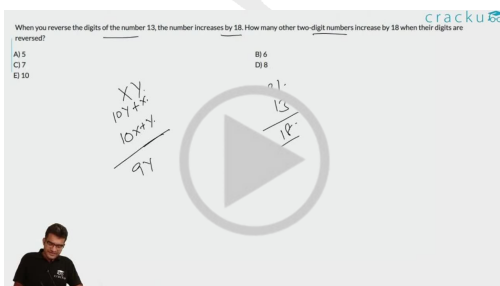
VIDEO SOLUTION

CAT Percentile Predictor



20. When you reverse the digits of the number 13, the number increases by 18. How many other two-digit numbers increase by 18 when their digits are reversed?

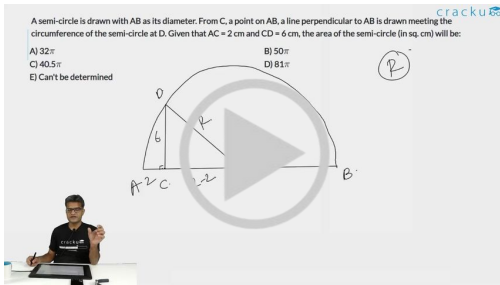
- A 5
- B 6
- C 7
- D 8
- E 10



VIDEO SOLUTION

21. A semi-circle is drawn with AB as its diameter. From C, a point on AB, a line perpendicular to AB is drawn meeting the circumference of the semi-circle at D. Given that AC = 2 cm and CD = 6 cm, the area of the semi-circle (in sq. cm) will be:

- A 32π
- B 50π
- C 40.5π
- D 81π
- E Can't be determined



VIDEO SOLUTION

22. There are 6 tasks and 6 persons. Task 1 cannot be assigned either to person 1 or to person 2; task 2 must be assigned to either person 3 or person 4. Every person is to be assigned one task. In how many ways can the assignment be done?

- A 144
- B 180
- C 192
- D 360
- E 716



VIDEO SOLUTION



23. The number of employees in Obelix Menhir Co. is a prime number and is less than 300. The ratio of the number of employees who are graduates and above, to that of employees who are not, can possibly be:

- A 101:88
- B 87:100
- C 110:111
- D 85:98
- E 97:84

24. If $\log_y x = (a * \log_z y) = (b * \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?

- A (-2, 1/2)
- B (1, 1)
- C (0.4, 2.5)
- D (π , $1/\pi$)
- E (2, 2)

 VIDEO SOLUTION

 VIDEO SOLUTION



27. Students who obtained Group Scores of at least 95 in every group are eligible to apply for a prize. Among those who are eligible, the student obtaining the highest Group Score in Social Science Group is awarded this prize. The prize was awarded to:

- A Shreya
- B Ram
- C Ayesha
- D Dipan
- E no one from the top ten

28. Among the top ten students, how many boys scored at least 95 in at least one paper from each of the groups?

- A 1
- B 2
- C 3
- D 4
- E 5

29. Each of the ten students was allowed to improve his/her score in exactly one paper of choice with the objective of maximizing his /her final score. Everyone scored 100 in the paper in which he or she chose to improve. After that, the topper among the ten students was:

- A Ram
- B Agni
- C Pritam
- D Ayesha
- E Dipan



30. Had Joseph, Agni, Pritam and Tirna each obtained Group Score of 100 in the Social Science Group, then their standing in decreasing order of final score would be;

- A Pritam, Joseph, Tirna, Agni
- B Joseph, Tirna, Agni, Pritam
- C Pritam, Agni, Tirna, Joseph
- D Joseph, Tirna, Pritam, Agni
- E Pritam, Tirna, Agni, Joseph

Instructions [31 - 35]

Mathematicians are assigned a number called Erdos number (named after the famous mathematician, Paul Erdos). Only Paul Erdos himself has an Erdos number of zero. Any mathematician who has written a research paper with Erdos has an Erdos number of 1. For other mathematicians, the calculation of his/her Erdos number is illustrated below:

Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y . Then X has an Erdos number of $y+1$. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

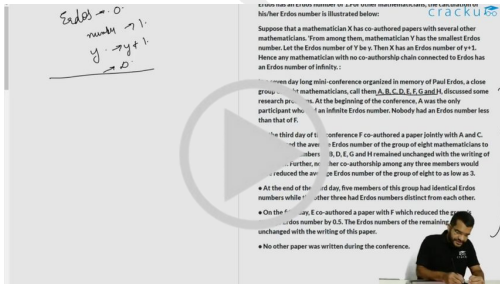
In a seven day long mini-conference organized in memory of Paul Erdos, a close group of eight mathematicians, call them A, B, C, D, E, F, G and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3. The Erdos numbers of B, D, E, G and H remained unchanged with the writing of this paper. Further, no other co-authorship among any three members would have reduced the average Erdos number of the group of eight to as low as 3.

- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fifth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining six were unchanged with the writing of this paper.
- No other paper was written during the conference.

31. The person having the largest Erdos number at the end of the conference must have had Erdos number (at that time):

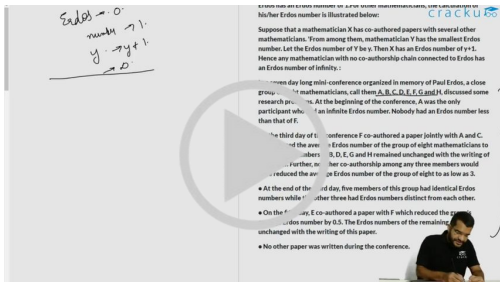
- A 5
- B 7
- C 9
- D 14
- E 15



VIDEO SOLUTION

32. How many participants in the conference did not change their Erdos number during the conference?

- A 2
- B 3
- C 4
- D 5
- E Cannot be determined

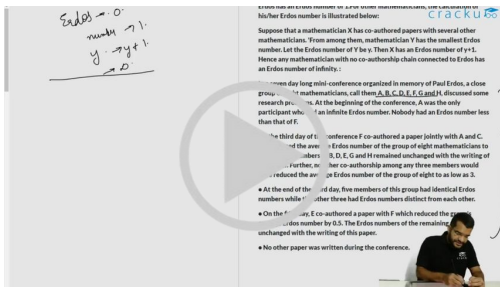


VIDEO SOLUTION



33. The Erdos number of C at the end of the conference was:

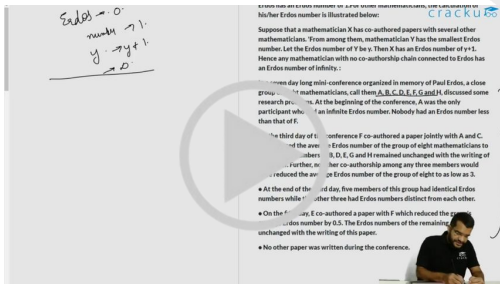
- A 1
- B 2
- C 3
- D 4
- E 5



VIDEO SOLUTION

34. The Erdos number of E at the beginning of the conference was:

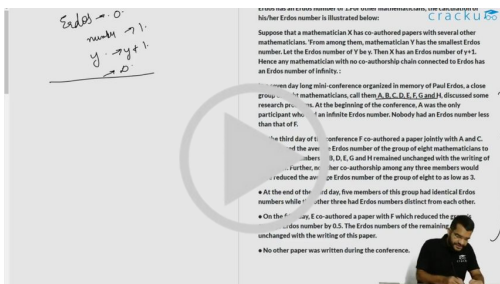
- A 2
- B 5
- C 6
- D 7
- E 8



VIDEO SOLUTION

35. How many participants had the same Erdos number at the beginning of the conference?

- A 2
- B 3
- C 4
- D 5
- E Cannot be determined



VIDEO SOLUTION



Instructions [36 - 40]

Two traders, Chetan and Michael, were involved in the buying and selling Of MCS shares over five trading days. At the beginning of the first day, the MCS share was priced at Rs 100, while at the end of the fifth day it was priced at Rs 110. At the end of each day, the MCS share price either went up by Rs 10, or else, it came down by Rs 10. Both Chetan and Michael took buying and selling decisions at the end of each trading day. The beginning price of MCS share on a given day was the same as the ending price of the previous day. Chetan and Michael started with the same number of shares and amount of cash, and had enough of both. Below are some additional facts about how Chetan and Michael traded over the five trading days.

- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price. On the other hand, each day if the price went down, he bought 10 shares at the closing price.
 - If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.
36. If Chetan sold 10 shares of MCS on three consecutive days, while Michael sold 10 shares only once during the five days, what was the price of MCS at the end of day 3?

- A Rs 90
- B Rs 100
- C Rs 110
- D Rs 120
- E Rs 130

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	110	110
5	110	120	90	100	110
6	110	100	110	100	110
7	110	100	110	110	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	130	110

decisions at the end of each trading day. The beginning price of MCS share on 1st day was Rs 100. Chetan and Michael started with the same number of shares and amount of cash, and had enough of both. Below are some additional facts about how Chetan and Michael traded over the five trading days.

- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price.
- On the other hand, each day if the price went down, he bought 10 shares at the closing price.
- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

VIDEO SOLUTION

37. If Michael ended up with Rs 100 less cash than Chetan at the end of day 5, what was the difference in the number of shares possessed by Michael and Chetan (at the end of day 5)?

- A Michael had 10 less shares than Chetan.
- B Michael had 10 more shares than Chetan.
- C Chetan had 10 more shares than Michael.
- D Chetan had 20 more shares than Michael.
- E Both had the same number of shares.

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	110	110
5	110	120	90	100	110
6	110	100	110	100	110
7	110	100	110	110	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	130	110

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- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price.
- On the other hand, each day if the price went down, he bought 10 shares at the closing price.
- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

VIDEO SOLUTION

38. If Chetan ended up with Rs 1300 more cash than Michael at the end of day 5, what was the price of MCS share at the end of day 4?

- A Rs 90
- B Rs 100
- C Rs 110
- D Rs 120
- E Not uniquely determinable

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	110	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	110	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	120	110

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- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price. On the other hand, each day if the price went down, he bought 10 shares at the closing price.
- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

VIDEO SOLUTION



39. What could have been the maximum possible increase in combined cash balance of Chetan and Michael at the end of the fifth day?

- A Rs 3700
- B Rs 4000
- C Rs 4700
- D Rs 5000
- E Rs 6000

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	110	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	110	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	120	110

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- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

VIDEO SOLUTION

40. If Michael ended up with 20 more shares than Chetan at the end of day 5, what was the price of the share at the end of day 3?

- A Rs 90
- B Rs 100
- C Rs 110
- D Rs 120
- E Rs 130

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	110	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	110	110
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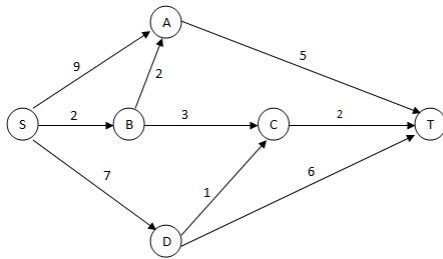
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VIDEO SOLUTION

Instructions [41 - 45]

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street. –



Motorists traveling from point S to point T would obviously take the route for which the total cost of traveling is the minimum. If two or more routes have the same least travel cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all the least cost routes.

The government can control the flow of traffic only by levying appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

41. If the government wants to ensure that all motorists travelling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from B to C is under repairs (and hence unusable), then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

- A 2,5,3,2
- B 0,5,3,2
- C 1,5,3,2
- D 2,3,5,1
- E 1,3,5,1

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street.

Motorists traveling from point S to point T would obviously take the route for which the total cost of traveling is the minimum. If two or more routes have the same least travel cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all the least cost routes.

The government can control the flow of traffic only by levying appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

If the government wants to ensure that all motorists travelling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from B to C is under repairs (and hence unusable), then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

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VIDEO SOLUTION

100+ CAT Quant Questions



42. If the government wants to ensure that no traffic flows on the street from D to T, while equal amount of traffic flows through junctions A and C, then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

- A 1,5,3,3
- B 1,4,4,3
- C 1,5,4,2
- D 0,5,2,3
- E 0,5,2,2

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street.

Motorists traveling from point S to point T would obviously take the route for which the total cost of traveling is the minimum. If two or more routes have the same least travel cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all the least cost routes.

The government can control the flow of traffic only by levying appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

If the government wants to ensure that all motorists travelling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from B to C is under repairs (and hence unusable), then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

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43. If the government wants to ensure that all routes from S to T get the same amount of traffic, then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

- A 0, 5, 2, 2
- B 0,5,4,1
- C 1,5,3,3
- D 1, 5, 3,2
- E 1,5,4,2

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street.

Motorists traveling from point S to point T would like to choose the route for which the total cost of traveling is the minimum. If two or more routes have the same total cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among the cost routes.

The government can control the flow of traffic or levy an appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

If the government wants to ensure that all motorists traveling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from B to C is under repairs (and hence unusable), then the toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

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VIDEO SOLUTION

44. If the government wants to ensure that the traffic at S gets evenly distributed along streets from S to A, from S to B, and from S to D, then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

- A 0,5,4,1
- B 0,5,2,2
- C 1,5,3,3
- D 1,5,3,2
- E 0,4,3,2

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street.

Motorists traveling from point S to point T would like to choose the route for which the total cost of traveling is the minimum. If two or more routes have the same total cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among the cost routes.

The government can control the flow of traffic or levy an appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

If the government wants to ensure that all motorists traveling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from B to C is under repairs (and hence unusable), then the toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

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45. The government wants to devise a toll policy such that the total cost to the commuters per trip is minimized. The policy should also ensure that not more than 70 per cent of the total traffic passes through junction B. The cost incurred by the commuter travelling from point S to point T under this policy will be:

- A Rs 7
- B Rs 9
- C Rs 10
- D Rs 13
- E Rs 14

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street.

Motorists travelling from point S to point T would like to know the route for which the total cost of travelling is the minimum. If two or more routes have the same total cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among the minimum cost routes.

The government can control the flow of traffic or levy appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

If the government wants to ensure that all motorists travelling from S to T pay the same amount (fuel costs and toll combined) for the route they choose and the street from B to C is under repairs (and hence unusable), then the toll charged (in rupees) at junctions A, C, and D respectively to achieve this goal is:

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VIDEO SOLUTION

Instructions [46 - 50]

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among P, R and S.
2. A team must include either M or Q, but not both.
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U and W, then it should also include the other two.
5. L and N cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in the team.

46. What could be the size of a team that includes K?

- A 2 or 3
- B 2 or 4
- C 3 or 4
- D Only 2
- E Only 4

VIDEO SOLUTION

47. In how many ways a team can be constituted so that the team includes N?

- A 2
- B 3
- C 4
- D 5
- E 6

100+ CAT VARC Questions



48. What would be the size of the largest possible team?

- A 8
- B 7
- C 6
- D 5
- E cannot be determined

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among R and S.
2. A team must include either M or Q but not both.
3. If a team includes L, then it must also include U, and vice versa.
4. If a team includes one among L, U and W, then it should also include the other two.
5. L and U cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in it.

Handwritten notes on the screen:

- For condition 1: R | S
- For condition 2: M | Q
- For condition 3: L | U
- For condition 4: L, U, W
- For condition 5: L, U
- For condition 6: L, U

49. Who can be a member of a team of size 5?

- A K
- B L
- C M
- D P
- E R

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among R and S.
2. A team must include either M or Q but not both.
3. If a team includes L, then it must also include U, and vice versa.
4. If a team includes one among L, U and W, then it should also include the other two.
5. L and U cannot be members of the same team.
6. L and U cannot be members of the same team.

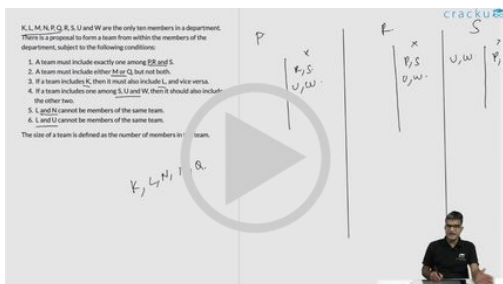
The size of a team is defined as the number of members in it.

Handwritten notes on the screen:

- For condition 1: R | S
- For condition 2: M | Q
- For condition 3: L | U
- For condition 4: L, U, W
- For condition 5: L, U
- For condition 6: L, U

50. Who cannot be a member of a team of size 3?

- A L
- B M
- C N
- D P
- E Q



VIDEO SOLUTION

CAT Expected Questions PDF



Verbal

Instructions [51 - 55]

Fifteen years after communism was officially pronounced dead, its spectre seems once again to be haunting Europe. Last month, the Council of Europe's parliamentary assembly voted to condemn the "crimes of totalitarian communist regimes," linking them with Nazism and complaining that communist parties are still "legal and active in some countries." Now Goran Lindblad, the conservative Swedish MP behind the resolution, wants to go further. Demands that European Ministers launch a continent-wide anti-communist campaign – including school textbook revisions, official memorial days, and museums – only narrowly missed the necessary two-thirds majority. Mr. Lindblad pledged to bring the wider plans back to the Council of Europe in the coming months:

He has chosen a good year for his ideological offensive: this is the 50th anniversary of Nikita Khrushchev's denunciation of Josef Stalin and the subsequent Hungarian uprising, which will doubtless be the cue for further excoriation of the communist record. Paradoxically, given that there is no communist government left in Europe outside Moldova, the attacks have if anything, become more extreme as time has gone on. A clue as to why that might be can be found in the rambling report by Mr. Lindblad that led to the Council of Europe declaration. Blaming class struggle and public ownership, he explained "different elements of communist ideology such as equality or social justice still seduce many" and "a sort of nostalgia for communism is still alive." Perhaps the real problem for Mr. Lindblad and his right-wing allies in Eastern Europe is that communism is not dead enough – and they will only be content when they have driven a stake through its heart.

The fashionable attempt to equate communism and Nazism is in reality a moral and historical nonsense. Despite the cruelties of the Stalin terror, there was no Soviet Treblinka or Sobibor, no extermination camps built to murder millions. Nor did the Soviet Union launch the most devastating war in history at a cost of more than 50 million lives – in fact it played the decisive role in the defeat of the German war machine. Mr. Lindblad and the Council of Europe adopt as fact the wildest estimates of those "killed by communist regimes" (mostly in famines) from the fiercely contested Black Book of Communism, which also underplays the number of deaths attributable to Hitler. But, in any case, none of this explains why anyone might be nostalgic in former communist states, now enjoying the delights of capitalist restoration.

The dominant account gives no sense of how communist regimes renewed themselves after 1956 or why Western leaders feared they might overtake the capitalist world well into the 1960s. For all its brutalities and failures, communism in the Soviet Union, Eastern Europe, and elsewhere delivered rapid industrialization, mass education, job security, and huge advances in social and gender equality. Its existence helped to drive up welfare standards in the west, and provided a powerful counterweight to western global domination.

It would be easier to take the Council of Europe's condemnation of communist state crimes seriously if it had also seen fit to denounce the far bloodier record of European colonialism – which only finally came to an end in the 1970s. This was a system of racist despotism, which dominated the globe in Stalin's time. And while there is precious little connection between the ideas of fascism and communism, there is an intimate link between colonialism and Nazism. The terms *lebensraum* and *konzentrationslager* were both first used by the German colonial regime in South-West Africa (now Namibia), which committed genocide against the Herero and Nama peoples and bequeathed its ideas and personnel directly to the Nazi parry.

Around 10 million Congolese died as a result of Belgian forced labor and mass murder in the early twentieth century; tens of millions perished in avoidable or enforced famines in British-ruled India; up to a million Algerians died in their war for independence, while controversy now rages in France about a new law requiring teachers to put a positive spin on colonial history. Comparable atrocities were carried out by all European colonialists, but not a word of condemnation from the Council of Europe. Presumably, European lives count for more.

No major twentieth century political tradition is without blood on its hands, but battles over history are more about the future than the past. Part of the current enthusiasm in official Western circles for dancing on the grave of communism is no doubt about relations with today's Russia and China. But it also reflects a determination to prove there is no alternative to the new global capitalist order – and that any attempt to find one is bound to lead to suffering. With the new imperialism now being resisted in the Muslim world and Latin America, growing international demands for social justice and ever greater doubts about whether the environmental crisis can be solved within the existing economic system, the pressure for alternatives will increase.

51. Among all the apprehensions that Mr. Goran Lindblad expresses against communism, which one gets admitted, although indirectly, by the author?

- A There is nostalgia for communist ideology even if communism has been abandoned by most European nations.
- B Notions of social justice inherent in communist ideology appeal to critics of existing systems.
- C Communist regimes were totalitarian and marked by brutalities and large scale violence.
- D The existing economic order is wrongly viewed as imperialistic by proponents of communism.
- E Communist ideology is faulted because communist regimes resulted in economic failures.

Fifteen years after communism was officially pronounced dead, its spectre seems once again to be haunting Europe. Last month, the Council of Europe's parliamentary assembly voted to condemn the "crimes of totalitarian communist regimes," linking them with Nazism and complaining that communist parties are still "legal and active in some countries." Now Goran Lindblad, the conservative Swedish MP behind the resolution, wants to go further. Demands that European Ministers launch a continent-wide anti-communist campaign – including school textbook revisions, official memorial days, and museums – only narrowly missed the necessary two-thirds majority. Mr. Lindblad pledged to bring the wider plans before the Council of Europe in the coming months. He has chosen a good year for his ideological offensive: this is the 50th anniversary of Nikita Khrushchev's denunciation of Josef Stalin and the subsequent Hungarian uprising, which will

VIDEO SOLUTION

CAT Important Topics

52. What, according to the author, is the real reason for a renewed attack against communism?

- A Disguising the unintended consequences of the current economic order such as social injustice and environmental crisis.
- B Idealising the existing ideology of global capitalism.
- C Making communism a generic representative of all historical atrocities, especially those perpetrated by the European imperialists.
- D Communism still survives, in bits and pieces, in the minds and hearts of people.
- E Renewal of some communist regimes has led to the apprehension that communist nations might overtake the capitalists.

Fifteen years after communism was officially pronounced dead, its spectre seems once again to be haunting Europe. Last month, the Council of Europe's parliamentary assembly voted to condemn the "crimes of totalitarian communist regimes," linking them with Nazism and complaining that communist parties are still "legal and active in some countries." Now Goran Lindblad, the conservative Swedish MP behind the resolution, wants to go further. Demands that European Ministers launch a continent-wide anti-communist campaign – including school textbook revisions, official memorial days, and museums – only narrowly missed the necessary two-thirds majority. Mr. Lindblad pledged to bring the wider plans before the Council of Europe in the coming months. He has chosen a good year for his ideological offensive: this is the 50th anniversary of Nikita Khrushchev's denunciation of Josef Stalin and the subsequent Hungarian uprising, which will

VIDEO SOLUTION

53. The author cites examples of atrocities perpetrated by European colonial regimes in order to

- A compare the atrocities committed by colonial regimes with those of communist regimes.
- B prove that the atrocities committed by colonial regimes were more than those of communist regimes.
- C prove that, ideologically, communism was much better than colonialism and Nazism.
- D Neutralise the arguments of Mr. Lindblad and to point out that the atrocities committed by colonial regimes were more than those of communist regimes.
- E neutralize the arguments of Mr. Lindblad and to argue that one needs to go beyond and look at the motives of these regimes.

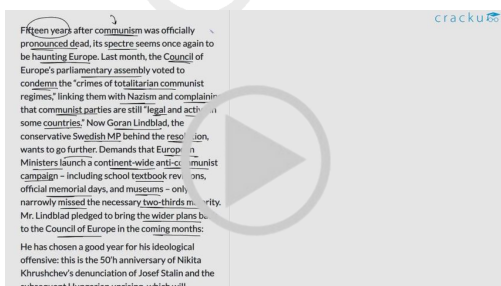
Fifteen years after communism was officially pronounced dead, its spectre seems once again to be haunting Europe. Last month, the Council of Europe's parliamentary assembly voted to condemn the "crimes of totalitarian communist regimes," linking them with Nazism and complaining that communist parties are still "legal and active in some countries." Now Goran Lindblad, the conservative Swedish MP behind the resolution, wants to go further. Demands that European Ministers launch a continent-wide anti-communist campaign – including school textbook revisions, official memorial days, and museums – only narrowly missed the necessary two-thirds majority. Mr. Lindblad pledged to bring the wider plans before the Council of Europe in the coming months. He has chosen a good year for his ideological offensive: this is the 50th anniversary of Nikita Khrushchev's denunciation of Josef Stalin and the subsequent Hungarian uprising, which will

VIDEO SOLUTION

54. Why, according to the author, is Nazism closer to colonialism than it is to communism?

- A Both colonialism and Nazism were examples of tyranny of one race over another.
- B The genocides committed by the colonial and the Nazi regimes were of similar magnitude.

- C Several ideas of the Nazi regime were directly imported from colonial regimes.
- D Both colonialism and Nazism are based on the principles of imperialism.
- E While communism was never limited to Europe, both the Nazis and the colonialists originated in Europe.

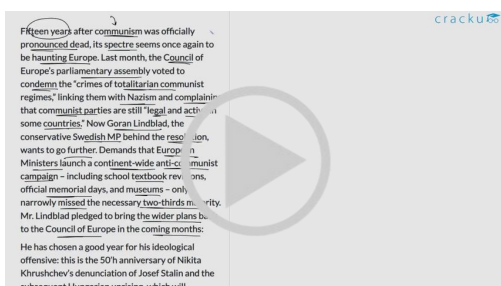


VIDEO SOLUTION



55. Which of the following cannot be inferred as a compelling reason for the silence of the Council of Europe on colonial atrocities?

- A The Council of Europe being dominated by erstwhile colonialists.
- B Generating support for condemning communist ideology.
- C Unwillingness to antagonize allies by raking up an embarrassing past.
- D Greater value seemingly placed on European lives.
- E Portraying both communism and Nazism as ideologies to be condemned.



VIDEO SOLUTION

Instructions [56 - 60]

My aim is to present a conception of justice which generalizes and carries to a higher level of abstraction the familiar theory of the social contract. In order to do this we are not to think of the original contract as one to enter a particular society or to set up a particular form of government. Rather, the idea is that the principles of justice for the basic structure of society are the object of the original agreement. They are the principles that free and rational persons concerned to further their own interests would accept in an initial position of equality. These principles are to regulate all further agreements; they specify the kinds of social cooperation that can be entered into and the forms of government that can be established. This way of regarding the principles of justice, I shall call justice as fairness. Thus, we are to imagine that those who engage in social cooperation choose together, in one joint act, the principles which are to assign basic rights and duties and to determine the division of social benefits. Just as each person must decide by rational reflection what constitutes his good, that is, the system of ends which it is rational for him to pursue, so a group of persons must decide once and for all what is to count among them as just and unjust. The choice which rational men would make in this hypothetical situation of equal liberty determines the principles of justice.

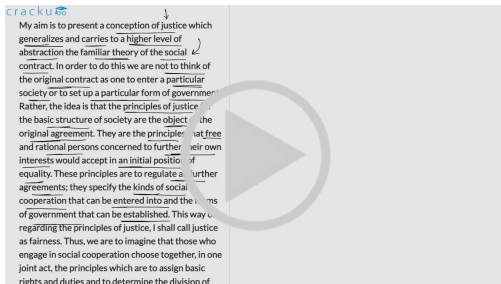
In 'justice as fairness', the original position is not an actual historical state of affairs. It is understood as a purely hypothetical situation characterized so as to lead to a certain conception of justice. Among the essential features of this situation is that no one knows his place in society, his class position or social status, nor does anyone know his fortune in the distribution of natural assets and abilities, his intelligence, strength, and the like. I shall even assume that the parties do not know their conceptions of the good or their special psychological propensities. The principles of justice are chosen behind a veil of ignorance. This ensures that no one is advantaged or disadvantaged in the choice of principles by the outcome of natural chance or the contingency of social circumstances. Since all are similarly situated and no one is able to design principles to favor his particular condition, the principles of justice are the result of a fair agreement or bargain.

Justice as fairness begins with one of the most general of all choices which persons might make together, namely, with the choice of the first principles of a conception of justice which is to regulate all subsequent criticism and reform of institutions. Then, having chosen a conception of justice, we can suppose that they are to choose a constitution and a legislature to enact laws, and so on, all in accordance with the principles of justice initially agreed upon. Our social situation is just if it is such that by this sequence of hypothetical agreements we would have contracted into the general system of rules which defines it. Moreover, assuming that the original position does determine a set of principles, it will then be true that whenever social institutions satisfy these principles, those engaged in them can say to one another that they are cooperating on terms to which they would agree if they were free and equal persons whose relations with respect to one another were fair. They could all view their arrangements as meeting the stipulations which they would acknowledge in an initial situation that embodies widely accepted and reasonable constraints on the choice of principles. The general recognition of this fact would provide the basis for a public acceptance of the corresponding principles of justice. No society can, of course, be a scheme of cooperation which men enter voluntarily in a literal

sense; each person finds himself placed at birth in some particular position in some particular society, and the nature of this position materially affects his life prospects. Yet a society satisfying the principles of justice as fairness comes as close as a society can to being a voluntary scheme, for it meets the principles which free and equal persons would assent to under circumstances that are fair.

56. A just society, as conceptualized in the passage, can be best described as:

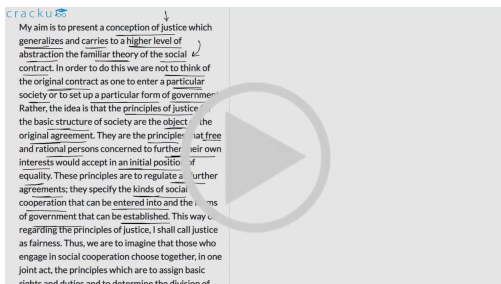
- A A Utopia in which everyone is equal and no one enjoys any privilege based on their existing positions and powers. '
- B A hypothetical society in which people agree upon principles of justice which are fair.
- C A society in which principles of justice are not based on the existing positions and powers of the individuals.
- D A society in which principles of justice are fair to all.
- E A hypothetical society in which principles of justice are not based on the existing positions and powers of the individuals.



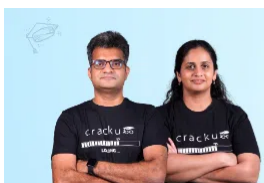
VIDEO SOLUTION

57. The original agreement or original position in the passage has been used by the author as:

- A A hypothetical situation conceived to derive principles of justice which are not influenced by position, status and condition of individuals in the society.
- B A hypothetical situation in which every individual is equal and no individual enjoys any privilege based on the existing positions and powers.
- C A hypothetical situation to ensure fairness of agreements among individuals in society.
- D An imagined situation in which principles of justice would have to be fair.
- E An imagined situation in which fairness is the objective of the principles of justice to ensure that no individual enjoys any privilege based on the existing positions and powers.



VIDEO SOLUTION



CAT Mock Test

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58. Which of the following best illustrates the situation that is equivalent to choosing 'the principles of justice' behind a 'veil of ignorance'?

- A The principles of justice are chosen by businessmen, who are marooned on an uninhabited island after a shipwreck, but have some possibility of returning.
- B The principles of justice are chosen by a group of school children whose capabilities are yet to develop.
- C The principles of justice are chosen by businessmen, who are marooned on an uninhabited island after a shipwreck and have no possibility of returning.
- D The principles of justice are chosen assuming that such principles will govern the lives of the rule makers only in their next birth if the rule makers agree that they will be born again.
- E The principles of justice are chosen by potential immigrants who are unaware of the resources necessary to succeed in a foreign country.

cracku

My aim is to present a conception of justice which generalizes and carries to a higher level of abstraction the familiar theory of the social contract. In order to do this we are not to think of the original contract as one to enter a particular society or to set up a particular form of government. Rather, the idea is that the principles of justice the basic structure of society are the object of the original agreement. They are the principles that free and rational persons concerned to further their own interests would accept in an initial position of equality. These principles are to regulate a further agreements; they specify the kinds of social cooperation that can be entered into and the terms of government that can be established. This way of regarding the principles of justice, I shall call justice as fairness. Thus, we are to imagine that those who engage in social cooperation choose together, in one joint act, the principles which are to assign basic rights and duties and to determine the division of

VIDEO SOLUTION

59. Why, according to the passage, do principles of justice need to be based on an original agreement?

- A Social institutions and laws can be considered fair only if they conform to principles of justice.
- B Social institutions and laws can be fair only if they are consistent with the principles of justice as initially agreed upon.
- C Social institutions and laws need to be fair in order to be just.
- D Social institutions and laws evolve fairly only if they are consistent with the principles of justice as initially agreed upon.
- E Social institutions and laws conform to the principles of justice as initially agreed upon.

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VIDEO SOLUTION

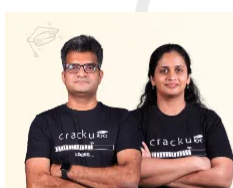
60. Which of the following situations best represents the idea of justice as fairness, as argued in the passage?

- A All individuals are paid equally for the work they do.
- B Everyone is assigned some work for his or her livelihood.
- C All acts of theft are penalized equally.
- D All children are provided free education in similar schools.
- E All individuals are provided a fixed sum of money to take care of their health.

cracku

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cracku



Instructions [61 - 65]

Our propensity to look out for regularities, and to impose laws upon nature, leads to the psychological phenomenon of dogmatic thinking or, more generally, dogmatic behaviour: we expect regularities everywhere and attempt to find them even where there are none; events which do not yield to these attempts we are inclined to treat as a kind of 'background noise'; and we stick to our expectations even when they are inadequate and we ought to accept defeat. This dogmatism is to some extent necessary. It is demanded by a situation which can only be dealt with by forcing our conjectures upon the world. Moreover, this dogmatism allows us to approach a good theory in stages, by way of approximations: if we accept defeat too easily, we may prevent ourselves from finding that we were very nearly right.

It is clear that this dogmatic attitude; which makes us stick to our first impressions, is indicative of a strong belief; while a critical attitude, which is ready to modify its tenets, which admits doubt and demands tests, is indicative of a weaker belief. Now according to Hume's theory, and to the popular theory, the strength of a belief should be a product of repetition; thus it should always grow with experience, and always be greater in less primitive persons. But dogmatic thinking, an uncontrolled wish to impose regularities, a manifest pleasure in rites and in repetition as such, is characteristic of primitives and children; and increasing experience and maturity sometimes create an attitude of caution and criticism rather than of dogmatism.

My logical criticism of Hume's psychological theory, and the considerations connected with it, may seem a little removed from the field of the philosophy of science. But the distinction between dogmatic and critical thinking, or the dogmatic and the critical attitude, brings us right back to our central problem. For the dogmatic attitude is clearly related to the tendency to verify our laws and schemata by seeking to apply them and to confirm them, even to the point of neglecting refutations, whereas the critical attitude is one of readiness to change them - to test them; to refute them; to falsify them, if possible. This suggests that we may identify the critical attitude with the scientific attitude, and the dogmatic attitude with the one which we have described as pseudo-scientific. It further suggests that genetically speaking the pseudo-scientific attitude is more primitive than, and prior to, the scientific attitude: that it is a pre-scientific attitude. And this primitivity or priority also has its logical aspect. For the critical attitude is not so much opposed to the dogmatic attitude as super-imposed upon it: criticism must be directed against existing and influential beliefs in need of critical revision - in other words, dogmatic beliefs. A critical attitude needs for its raw material, as it were, theories or beliefs which are held more or less dogmatically.

Thus, science must begin with myths, and with the criticism of myths; neither with the collection of observations, nor with the invention of experiments, but with the critical discussion of myths, and of magical techniques and practices. The scientific tradition is distinguished from the pre-scientific tradition in having two layers. Like the latter, it passes on its theories; but it also passes on a critical attitude towards them. The theories are passed on, not as dogmas, but rather with the challenge to discuss them and improve upon them.

The critical attitude, the tradition of free discussion of theories with the aim of discovering their weak spots so that they may be improved upon, is the attitude of reasonableness, of rationality. From the point of view here developed, all laws, all theories, remain essentially tentative, or conjectural, or hypothetical, even when we feel unable to doubt them any longer. Before a theory has been refuted we can never know in what way it may have to be modified.

61. In the context of science, according to the passage, the interaction of dogmatic beliefs and critical attitude can be best described as:

- A A duel between two warriors in which one has to die.
- B The effect of a chisel on a marble stone while making a sculpture.
- C The feedstock (natural gas) in fertilizer industry being transformed into fertilizers.
- D A predator killing its prey.
- E The effect of fertilizers on a sapling.

Our propensity to look out for regularities, and to impose laws upon nature, leads to the psychological phenomenon of dogmatic thinking or, more generally, dogmatic behaviour: we expect regularities everywhere and attempt to find them even where there are none; events which do not yield to these attempts we are inclined to treat as a kind of 'background noise'; and we stick to our expectations even when they are inadequate and we ought to accept defeat. This dogmatism is to some extent necessary. It is demanded by a situation which can only be dealt with by forcing our conjectures upon the world. Moreover, this dogmatism allows us to approach a good theory in stages, by way of approximations: if we accept defeat too easily, we may prevent ourselves from finding that we were very nearly right.

It is clear that this dogmatic attitude; which makes us stick to our first impressions, is indicative of a strong belief; while a critical attitude, which is ready to modify its tenets, which admits doubt and demands

VIDEO SOLUTION

62. According to the passage, the role of a dogmatic attitude of dogmatic behaviour in the development of science is

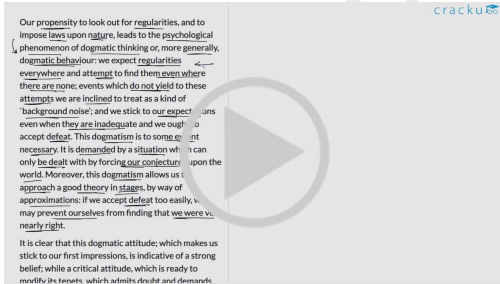
- A critical and important, as, without it, initial hypotheses or conjectures can never be made.
- B positive, as conjectures arising out of our dogmatic attitude become science.
- C negative, as it leads to pseudo-science.
- D neutral, as the development of science is essentially because of our critical attitude.
- E inferior to critical attitude, as a critical attitude leads to the attitude of reasonableness and rationality.

Our propensity to look out for regularities, and to impose laws upon nature, leads to the psychological phenomenon of dogmatic thinking or, more generally, dogmatic behaviour: we expect regularities everywhere and attempt to find them even where there are none; events which do not yield to these attempts we are inclined to treat as a kind of 'background noise'; and we stick to our expectations even when they are inadequate and we ought to accept defeat. This dogmatism is to some extent necessary. It is demanded by a situation which can only be dealt with by forcing our conjectures upon the world. Moreover, this dogmatism allows us to approach a good theory in stages, by way of approximations: if we accept defeat too easily, we may prevent ourselves from finding that we were very nearly right.

It is clear that this dogmatic attitude; which makes us stick to our first impressions, is indicative of a strong belief; while a critical attitude, which is ready to modify its tenets, which admits doubt and demands

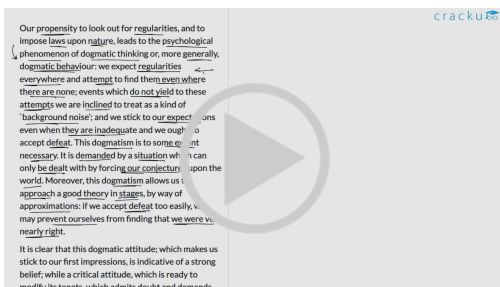
63. Dogmatic behaviour, in this passage, has been associated with primitives and children. Which of the following best describes the reason why the author compares primitives with children?

- A Primitives are people who are not educated, and hence can be compared with children, who have not yet been through school.
- B Primitives are people who, though not modern, are as innocent as children.
- C Primitives are people without a critical attitude, just as children are.
- D Primitives are people in the early stages of human evolution; similarly, children are in the early stages of their lives.
- E Primitives are people who are not civilized enough, just as children are not.



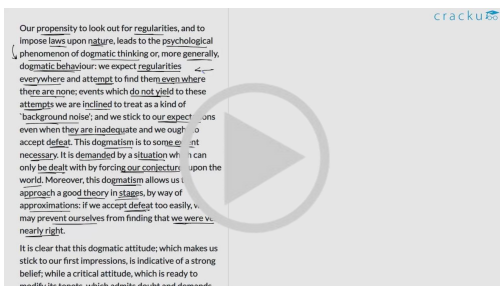
64. Which of the following statements best supports the argument in the passage that a critical attitude leads to a weaker belief than a dogmatic attitude does?

- A A critical attitude implies endless questioning, and, therefore, it cannot lead to strong beliefs.
- B A critical attitude, by definition, is centered on an analysis of anomalies and "noise".
- C A critical attitude leads to questioning everything, and in the process generates "noise" without any conviction.
- D A critical attitude is antithetical to conviction, which is required for strong beliefs.
- E A critical attitude leads to questioning and to tentative hypotheses.



65. According to the passage, which of the following statements best describes the difference between science and pseudo-science?

- A Scientific theories or hypothesis are tentatively true whereas pseudo-sciences are always true.
- B Scientific laws and theories are permanent and immutable whereas pseudo-sciences are contingent on the prevalent mode of thinking in a society.
- C Science always allows the possibility of rejecting a theory or hypothesis, whereas pseudo-sciences seek to validate their ideas or theories.
- D Science focuses on anomalies and exceptions so that fundamental truths can be uncovered, whereas pseudo-sciences focus mainly on general truths.
- E Science progresses by collection of observations or by experimentation, whereas pseudo-sciences do not worry about observations and experiments.



VIDEO SOLUTION

66. Facts, which deal with pieces of information that one has heard, seen or read, and which are open to discovery or verification (the answer option indicates such a statement with an 'F').

Inferences, which are conclusions drawn about the unknown, on the basis of the known (the answer option indicates such a statement with an 'I').

Judgements which are opinions that imply approval or disapproval of persons, objects, situations and occurrences in the past, the present or the future (the answer option indicates such a statement with a J)

Select the answer option that best describes the set of four statements.

1. So much of our day-to-day focus seems to be on getting things done, trudging our way through the tasks of living - it can feel like a treadmill that gets you nowhere; where is the childlike joy?
2. We are not doing the things that make us happy; that which brings us joy; the things that we cannot wait to do because we enjoy them so much.
3. This is the stuff that joyful living is made of - identifying your calling and committing yourself wholeheartedly to it.
4. When this happens, each moment becomes a celebration of you; there is a rush of energy that comes with feeling completely immersed in doing what you love most.

- A IIIJ
- B IFIJ
- C JFJJ
- D JJJJ
- E JFII



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67. Facts, which deal with pieces of information that one has heard, seen or read, and which are open to discovery or verification (the answer option indicates such a statement with an 'F').

Inferences, which are conclusions drawn about the unknown, on the basis of the known (the answer option indicates such a statement with an 'I').

Judgements which are opinions that imply approval or disapproval of persons, objects, situations and occurrences in the past, the present or the future (the answer option indicates such a statement with a J)

Select the answer option that best describes the set of four statements.

1. Given the poor quality of service in the public sector, the HIV/AIDS affected should be switching to private initiatives that supply anti-retroviral drugs (ARVs) at a low cost.
2. The government has been supplying free drugs since 2004, and 35000 have benefited up to now - though the size of the affected population is 150 times this number.
3. The recent initiatives of networks and companies like AIDSCare Network, Emcure, Reliance-Cipla-CII, would lead to availability of much-needed drugs to a larger number of affected people.
4. But how ironic it is that we should face a perennial shortage of drugs when India is one of the world's largest suppliers of generic drugs to the developing world.

- A JFIJ
- B JIIJ
- C IFIJ
- D IFFJ
- E JFII

68. Facts, which deal with pieces of information that one has heard, seen or read, and which are open to discovery or verification (the answer option indicates such a statement with an 'F').

Inferences, which are conclusions drawn about the unknown, on the basis of the known (the answer option indicates such a statement with an 'I').

Judgements which are opinions that imply approval or disapproval of persons, objects, situations and occurrences in the past, the present or the future (the answer option indicates such a statement with a J)

Select the answer option that best describes the set of four statements.

1. According to all statistical indications, the Sarva Shiksha Abhiyan has managed to keep pace with its ambitious goals.
2. The Mid-day Meal Scheme has been a significant incentive for the poor to send their little ones to school, thus establishing the vital link between healthy bodies and healthy minds.
3. Only about 13 million children in the age group of 6 to 14 years are out of school.
4. The goal of universalisation of elementary education has to be a pre-requisite for the evolution and development of our country.

- A IIFJ
- B JIIJ
- C IJFJ
- D IJFI
- E JIFT

69. Facts, which deal with pieces of information that one has heard, seen or read, and which are open to discovery or verification (the answer option indicates such a statement with an 'F').

Inferences, which are conclusions drawn about the unknown, on the basis of the known (the answer option indicates such a statement with an 'I').

Judgements which are opinions that imply approval or disapproval of persons, objects, situations and occurrences in the past, the present or the future (the answer option indicates such a statement with a J)

Select the answer option that best describes the set of four statements.

1. We should not be hopelessly addicted to an erroneous belief that corruption in India is caused by the crookedness of Indians.
2. The truth is that we have more red tape - we take eighty-nine days to start a small business, Australians take two.
3. Red tape leads to corruption and distorts a people's character.
4. Every red tape procedure is a point of contact with an official, and such contacts have the potential to become opportunities for money to change hands.

- A JFIF
- B JFJJ
- C JIJF
- D IFJF
- E JFJI



70. Facts, which deal with pieces of information that one has heard, seen or read, and which are open to discovery or verification (the answer option indicates such a statement with an 'F').

Inferences, which are conclusions drawn about the unknown, on the basis of the known (the answer option indicates such a statement with an 'I').

Judgements which are opinions that imply approval or disapproval of persons, objects, situations and occurrences in the past, the present or the future (the answer option indicates such a statement with a J)

Select the answer option that best describes the set of four statements.

1. Inequitable distribution of all kinds of resources is certainly one of the strongest and most sinister sources of conflict.
2. Even without war, we know that conflicts continue to trouble us - they only change in character.
3. Extensive disarmament is the only insurance for our future; imagine the amount of resources that can be released and redeployed.
4. The economies of the industrialized western world derive 20% of their income from the sale of all kinds of arms.

- A IJJI
- B JIJF
- C IJIF
- D JIIF
- E IJIF

71. From the given options, choose the sentence that completes the paragraph in the most appropriate way.

I am sometimes attacked for imposing 'rules'. Nothing could be further from the truth: I hate rules. All I do is report on how consumers react to different stimuli. I may say to a copywriter, "Research shows that commercials with celebrities are below average in persuading people to buy products. Are you sure you want to use a celebrity?" Call that a rule? Or I may say to an art director, "Research suggests that if you set the copy in black type on a white background, more people will read it than if you set it in white type on a black background."

- A Guidance based on applied research can hardly qualify as 'rules'.
- B Thus, all my so called 'rules' are rooted in applied research. .
- C A suggestion perhaps, but scarcely a rule.

- D Such principles are unavoidable if one wants to be systematic about consumer behaviour.
- E Fundamentally it is about consumer behaviour - not about celebrities or type settings.

72. From the given options, choose the sentence that completes the paragraph in the most appropriate way.

Relations between the factory and the dealer are distant and usually strained as the factory tries to force cars on the dealers to smooth out production. Relations between the dealer and the customer are equally strained because dealers continuously adjust prices - make deals - to adjust demand with supply while maximizing profits. This becomes a system marked by 'a lack of long-term commitment' on either side, which maximizes feelings of mistrust. In order to maximize their bargaining positions, everyone holds back information - the dealer about the product and the consumer about his true desires.

- A As a result, 'deal making' becomes rampant, without concern for customer satisfaction.
- B As a result, inefficiencies creep into the supply chain.
- C As a result, everyone treats the other as an adversary, rather than as an ally.
- D As a result, fundamental innovations are becoming scarce in the automobile industry.
- E As a result, everyone loses in the long run.



73. From the given options, choose the sentence that completes the paragraph in the most appropriate way.

In the evolving world order, the comparative advantage of the United States lies in its military force: Diplomacy and international law have always been regarded as annoying encumbrances, unless they can be used to advantage against an enemy. Every active player in world affairs professes to seek only peace and to prefer negotiation to violence and coercion.

- A However, diplomacy has often been used as a mask by nations which intended to use force.
- B However, when the veil is lifted, we commonly see that diplomacy is understood as a disguise for the rule of force.
- C However, history has shown that many of these nations do not practice what they profess.
- D However, history tells us that peace is professed by those who intend to use violence.
- E However, when unmasked, such nations reveal a penchant for the use of force.

74. From the given options, choose the sentence that completes the paragraph in the most appropriate way.

Age has a curvilinear relationship with the exploitation of opportunity. Initially, age will increase the likelihood that a person will exploit an entrepreneurial opportunity because people gather much of the knowledge necessary to exploit opportunities over the course of their lives, and because age provides credibility in transmitting that information to others. However, as people become older, their willingness to bear risks declines, their opportunity costs rise, and they become less receptive to new information.

[CAT 2006]

- A As a result, people transmit more information rather than experiment with new ideas as they reach an advanced age.
- B As a result, people are reluctant to experiment with new ideas as they reach an advanced age.
- C As a result, only people with lower opportunity costs exploit opportunity when they reach an advanced age.
- D As a result, people become reluctant to exploit entrepreneurial opportunities when they reach an advanced age.
- E As a result, people depend on credibility rather than on novelty as they reach an advanced age.

75. From the given options, choose the sentence that completes the paragraph in the most appropriate way.

We can usefully think of theoretical models as maps, which help us navigate unfamiliar territory. The most accurate map that it is possible to construct would be of no practical use whatsoever, for it would be an exact replica, on exactly the same scale, of the place where we were. Good maps pull out the most important features and throw away a huge amount of much less valuable information. Of course, maps can be bad as well as good - witness the attempts by medieval Europe to produce a map of the world. In the same way, a bad theory, no matter how impressive it may seem in principle, does little or nothing to help us understand a problem.

- A But good theories, just like good maps, are invaluable, even if they are simplified.
- B But good theories, just like good maps, will never represent unfamiliar concepts in detail.
- C But good theories, just like good maps, need to balance detail and feasibility of representation.
- D But good theories, just like good maps, are accurate only at a certain level of abstraction.
- E But good theories, just like good maps, are useful in the hands of a user who knows their limitations.



Answers

Quant

1. B	2. D	3. D	4. B	5. B	6. B	7. A	8. E
9. A	10. D	11. E	12. B	13. B	14. C	15. D	16. D
17. A	18. E	19. C	20. B	21. B	22. A	23. E	24. E
25. E							

Data Interpretation

26. C	27. D	28. A	29. E	30. A	31. B	32. D	33. B
34. C	35. B	36. C	37. E	38. B	39. D	40. A	41. C
42. E	43. D	44. A	45. C	46. E	47. E	48. D	49. C
50. A							

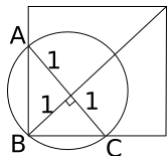
Verbal

51. C	52. B	53. E	54. A	55. D	56. C	57. A	58. D
59. B	60. D	61. B	62. A	63. D	64. E	65. C	66. D
67. A	68. C	69. E	70. B	71. C	72. E	73. B	74. D
75. A							

Explanations

Quant

1. **B**



The area of triangle ABC is $\frac{1}{2} * \sqrt{2} * \sqrt{2} = 1$

Area of semi-circle ABC = $\pi/2$

So, area of circle outside the square = $\pi/2 - 1 = (\pi - 2)/2$

So, area of circle inside the sheet = $\pi - (\pi/2 - 1) = 1 + \pi/2$

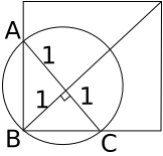
Area of original square = $2*2 = 4$

So, area of the sheet after punching = $4 - 1 - \pi/2 = 3 - \pi/2$

So, proportion of sheet that remains after punching = $(3 - \pi/2)/4 = (6 - \pi)/8$

VIDEO SOLUTION

2. D



The area of triangle ABC is $\frac{1}{2} * \sqrt{2} * \sqrt{2} = 1$

Area of semi-circle ABC = $\pi/2$

So, area of circle outside the square = $\pi/2 - 1 = (\pi - 2)/2$

VIDEO SOLUTION

Explanation [3 - 4]:

Let the limit be x and the rate of charge be k per kg.

Let the excess luggage with Raja be R kg.

So, excess luggage with Praja = 2R kg

Now, excess luggage with Raja + excess luggage with Praja = 60 - 2x

So, $3R = 60 - 2x \Rightarrow R = 20 - 2x/3$ which was charged 1200 Also, if one person had the entire luggage, excess luggage would have been 60 - x, which would have been charged 5400.

3. D

So the charge for the excess of $(20 - \frac{2x}{3}) = k(20 - \frac{2x}{3}) = 1200$ (1)

Also, the charge for the excess of $60 - x = k(60 - x) = 5400$ (2)

Dividing (1) by (2), we get

$$\Rightarrow \frac{(60 - 2x)}{3 \times (60 - x)} = \frac{1200}{5400}$$

Solving this, x = 15 kg

So, Praja's luggage = 35 kg

VIDEO SOLUTION

4. B

$$\Rightarrow (60 - 2x)/3 * (60 - x) = 1200/5400$$

Solving this, x = 15 kg

VIDEO SOLUTION

5. B

2^p is always positive

x^2 is always non negative.

$1/\sqrt{-x}$ is always positive.

$\frac{1}{x}$ is negative when x is negative.

In this case, x is negative $\Rightarrow \frac{1}{x}$ is smallest.

VIDEO SOLUTION



6. B

Make the power equal and compare the denominators.

$2^{1/2}$ can be written as $64^{1/12}$

$3^{1/3}$ can be written as $81^{1/12}$

$4^{1/4}$ can be written as $64^{1/12}$

$6^{1/6}$ can be written as $36^{1/12}$

Among these, $81^{1/12}$ is the greatest $\Rightarrow 3^{1/3}$ is the greatest.

VIDEO SOLUTION

7. A

$$a/d = a/b * b/c * c/d = 1/3 * 2 * 1/2 = 1/3$$

Similarly, b/e and c/f are 3 and 3/8 respectively.

$$b/e = b/c * c/d * d/e = 3$$

$$c/f = c/d * d/e * e/f = 3/8$$

$$\Rightarrow \text{Value of } abc/def = 1/3 * 3 * 3/8 = 3/8$$

VIDEO SOLUTION

8.E

The area of the four walls is length*height*2 + breadth*height*2

Initial area = $3*1*2 + 2*1*2 = 10$

Final area = $6*1/2*2 + 1*1/2*2 = 7$

So, the area decreased by 30%

9.A

substituting 3,4...53 in the given function, we get

$$\begin{aligned}t_3 &= \frac{3}{5} \\t_4 &= \frac{4}{5} \\t_5 &= \frac{5}{5} \\t_6 &= \frac{6}{8}\end{aligned}$$

Multiplying the values, we get $\frac{3}{5} * \frac{4}{6} * \frac{5}{7} * \dots * \frac{52}{54} * \frac{53}{55}$ which ultimately after cancellations give $\frac{3*4}{54*55} = \frac{2}{495}$

VIDEO SOLUTION

10.D

Let x be in the front row.

So no. of children in next rows will be x-3, x-6, x-9, x-12, x-15, x-18, x-21....

Suppose there are 6 rows, then the sum is equal to $x + x-3 + x-6 + x-9 + x-12 + x-15 = 6x - 45$

This sum is equal to 630.

$\Rightarrow 6x - 45 = 630 \Rightarrow 6x = 585$

Here, x is not an integer.

Hence, there cannot be 6 rows.

VIDEO SOLUTION



11.E

$$2^{0.7x} * 3^{-1.25y} = 8\sqrt{6}/27 \Rightarrow 2^{0.7x} * 3^{-1.25y} = 2^{3.5} * 3^{-2.5}$$

$$\Rightarrow 0.7x = 3.5 \Rightarrow x = 5$$

$$\Rightarrow -1.25y = -2.5 \Rightarrow y = 2$$

$$4^{0.3x} * 9^{0.2y} = 8 * 81^{1/5} \Rightarrow 2^{0.6x} * 3^{0.4y} = 2^3 * 3^{0.8}$$

$$\Rightarrow 0.6x = 3 \Rightarrow x = 5$$

$$\Rightarrow 0.4y = 0.8 \Rightarrow y = 2$$

$\Rightarrow (5,2)$ is the solution.

12.B

$$y = 38 \Rightarrow x = 1$$

$$y = 36 \Rightarrow x = 2$$

...

...

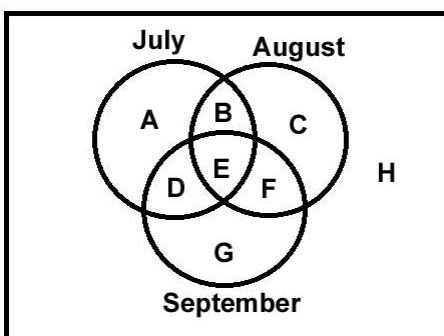
$$y = 14 \Rightarrow x = 13$$

$$y = 12 \Rightarrow x = 14 \Rightarrow \text{Cases from here are not valid as } x > y.$$

Hence, there are 13 solutions.

VIDEO SOLUTION

13.B



Let the areas be labelled as shown in the diagram above.

The number of people corresponding to "none of the three months" is 24. So, H is 24.

Only September is 18. So, G = 18

September but not August is 23. So, G + D = 23.

Hence, D = 23 - 18 = 5.

We know that September and July is 8. So, D + E = 8

This implies E = 3.

September = 28. So, D + E + F + G = 28.

So, F = 28 - 5 - 3 - 18 = 2.

July and August = 10.

So, B + E = 10.

E = 3. So, B = 7.

July = 48.

So, A + B + D + E = 48

A = 48 - 7 - 5 - 3 = 33.

There are 100 people in total. So, C = 100 - A - B - D - E - F - G - H = 100 - 33 - 7 - 5 - 3 - 2 - 18 - 24 = 8

So, number of people who read exactly two consecutive issues = (July & August) + (August & September) = B + F = 7 + 2 = 9

14. C

Maximum sum of the four numbers $\leq 384 = 99 + 97 + 95 + 93$

$384/10 = 38.4$

So, the perfect square is a number less than 38.4

The possibilities are 36, 25, 16 and 9

For the sum to be 360, the numbers can be 87, 89, 91 and 93

The sum of four consecutive odd numbers cannot be 250

For the sum to be 160, the numbers can be 37, 39, 41 and 43

The sum of 4 consecutive odd numbers cannot be 90

So, from the options, the answer is 41.

[VIDEO SOLUTION](#)

15. D

For a normal graph with y and x-axis, the equation of the line passing through the origin is $y = mx$ where m is the slope of the line.

m is +ve if the angle made by the line with the x-axis is $< 90^\circ$

$\therefore$ The equation of the line in the given graph would be $y - x = k(y + x)$ since the axes are y-x and y+x and the line is passing through the origin.

$k > 1$ because the angle is greater than 45°

$$y = \frac{x(k+1)}{1-k}$$

Since $k > 1$

Therefore $y < 0$ for $x > -1$ and $y > 0$ for $x < -1$

Option d correctly satisfy this condition



16. D

The nth term is $a + (n-1)d$

$1000 = 1 + (n-1)d$

So, $(n-1)d = 999$

$999 = 3^3 \times 37$

So, the number of factors is $4 \times 2 = 8$

Since there should be at least 3 terms in the series, d cannot be 999.

So, the number of possibilities is 7

[VIDEO SOLUTION](#)

17. A

Try to solve this type of questions using the options.

Substitute 0 first $\Rightarrow$ We get $-2 \leq 0$, which is correct. Hence, 0 must be in the solution set.

Substitute 8 $\Rightarrow 4 + 2 - 2 \leq 0 \Rightarrow 6 \leq 0$, which is false. Hence, 8 must not be in the solution set.

$\Rightarrow$ Option 1 is the answer.

[VIDEO SOLUTION](#)

18. E

The minimum value is obtained when $2x+1 = 3-4x \Rightarrow 6x = 2 \Rightarrow x = 1/3$

So, $f(x) = 2 \cdot 1/3 + 1 = 5/3$

VIDEO SOLUTION

19. C

Let the distance be D.

Time taken by Arun = $D/30$

Time taken by Barun = $D/40$

Now, $D/40 = D/30 - 2$

$\Rightarrow 3D = 4D - 240$

$\Rightarrow D = 240$

Therefore time taken by Arun to cover 240 km = $240/30 = 8$ hr

Time Kiranmala takes to cover 240 km = $240/60 = 4$ hr

So, Kiranmala has to start 4 hours after Arun.

VIDEO SOLUTION

20. B

Let the number be xy

$10y + x = 10x + y + 18$

$\Rightarrow 9y - 9x = 18$

$\Rightarrow y - x = 2$

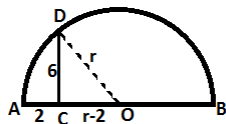
So, y can take values from 9 to 4 (since 3 is already counted in 13)

Number of possible values = 6

VIDEO SOLUTION



21. B



In triangle CDO, which is a right angled triangle, we can use pythagoras theorem.

$$6^2 + (r - 2)^2 = r^2$$

$$\Rightarrow 36 + r^2 - 4r + 4 = r^2$$

$$\Rightarrow 4r = 40$$

$$\Rightarrow r = 10$$

$$\Rightarrow \text{Area} = \pi * 10 * 10/2 = 50\pi$$

VIDEO SOLUTION

22. A

If the first task is assigned to either person 3 or person 4, the second task can be assigned in only 1 way. If the first task is assigned to either person 5 or person 6, the second task can be assigned in 2 ways. Therefore, the number of ways in which the first two tasks can be assigned is $2*1 + 2*2 = 6$.

The other 4 tasks can be assigned to 4 people in 4! ways.

The total number of ways of assigning the 6 tasks is, therefore, $6*4! = 144$.

VIDEO SOLUTION

23. E

The addition of numerator and denominator should give a prime no. Only option E gives that.

3 is a factor of 189 and 183 $\Rightarrow$ A and D eliminated

17 is a factor of 187 and 221 $\Rightarrow$ B and C eliminated

181 is prime.

24. E

$$\log_y x = ab$$

$$a * \log_z y = ab \Rightarrow \log_z y = b$$

$$b * \log_x z = ab \Rightarrow \log_x z = a$$

$$\log_y x = \log_z y * \log_x z \Rightarrow \log x / \log y = \log y / \log z * \log z / \log x$$

$$\Rightarrow \frac{\log x}{\log y} = \frac{\log y}{\log x}$$

$$\Rightarrow (\log x)^2 = (\log y)^2$$

$$\Rightarrow \log x = \log y \text{ or } \log x = -\log y$$

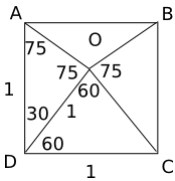
So, $x = y$ or $x = 1/y$

So, $ab = 1$ or -1

Option 5) is not possible

VIDEO SOLUTION

25. E



Triangle AOD is isosceles. So, angle DAO = angle DOA = 75. Similarly, angle BOC = 75. So, angle AOB = 150

VIDEO SOLUTION



Data Interpretation

26. C

Since we know the average, we can calculate the marks Dipan got in English Paper II (x) as ; $98+95+95.5+95+(96+x)/2 = 96*5$. We get $96+x = 96.5*2$. Thus $x = 97$. Hence option C.

27. D

From the given data it is clear that Dipan is the only student who obtained Group Scores of at least 95 in every group and obtained the highest Group Score in Social Science Group. Hence option D.

28. A

Among the top ten students, only Dipan scored at least 95 in at least one paper from each of the groups. Hence option A.

29. E

Lets first consider ayesha's marks, her 7 marks will increase in geography. So increase in her average marks is $7/(2*5) = 0.7$. So new average is 96.9. Now for dipan, his 5 marks will increase in mathematics. So increase in his average $5/5 = 1$. So his new average is $96+1 = 97$ which is greater than ayesha's. Other will clearly have a lower average than these both. Hence option E.

30. A

Had Joseph, Agni, Pritam and Tirna each obtained Group Score of 100 in the Social Science Group, their final score would increase by 0.9, 0.9, 2.2, 2.1 respectively. Adding these to the final score then their standing in decreasing order of final score would be Pritam, Joseph, Tirna, Agni with scores of 96.1, 95.9, 95.8, 95.2 respectively. Hence option A.



Explanation [31 - 35]:

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number, the values of c,a will change to f+1 and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos number then the Erdos number of co-authors will be min Erdos value + 1]

31. B

Average of the mathematicians is 3

Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day:f+1,b,f+1,d,e,f,g,h

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged.

Sum of the Erdos numbers of eight mathematicians=20

So here the difference of $4[24-20]$ arose, which means e will be f+5 initially and changed to f+1 after co-authoring with F.

So the Erdos number at the third day:f+1,b,f+1,d,f+5,f,g,h

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be f+5 because then there will be two mathematicians with the same Erdos number f+1.

So five mathematicians will have f+1, one with f+5, one with f, one with some different value say x

$$5(f+1)+f+5+f+x=24$$

$$7f+x=14$$

The only value which satisfies the above equation is f=1,x=7

Erdos number at the end of fifth day,f+1,b,f+1,d,f+1,f,g,h

On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	f+1	f+1
B	b	b	b
C	c	f+1	f+1
D	d	d	d
E	e	e = f+5	f+1
F	f(min)	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Hence the person having the largest Erdos number at the end of the conference must have had Erdos number 7 . Hence option B.

[VIDEO SOLUTION](#)

32. D

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day:f+1,b,f+1,d,e,f,g,h

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged. Sum of the Erdos numbers of eight mathematicians=20 So here the difference of 4[24-20] arose, which means e will be f+5 initially and changed to f+1 after co-authoring with F.

So the Erdos number at the third day:f+1,b,f+1,d,f+5,f,g,h

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be f+5 because then there will be two mathematicians with the same Erdos number f+1.

So five mathematicians will have f+1, one with f+5, one with f, one with some different value say x

$$5(f+1)+f+5+f+x=24$$

$$7f+x=14$$

The only value which satisfies the above equation is f=1,x=7

Erdos number at the end of fifth day,f+1,b,f+1,d,f+1,f,g,h On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	f+1	f+1
B	b	b	b
C	c	f+1	f+1
D	d	d	d
E	e	e = f+5	f+1
F	f(min)	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

So B,D ,F,G,H are 5 participants in the conference who did not change their Erdos number during the conference.

[VIDEO SOLUTION](#)

33. B

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day:f+1,b,f+1,d,e,f,g,h

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged. Sum of the Erdos numbers of eight mathematicians=20 So here the difference of 4[24-20] arose, which means e will be f+5 initially and changed to f+1 after co-authoring with F.

So the Erdos number at the third day:f+1,b,f+1,d,f+5,f,g,h

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be f+5 because then there will be two mathematicians with the same Erdos number f+1.

So five mathematicians will have f+1, one with f+5, one with f, one with some different value say x

$$5(f+1)+f+5+f+x=24$$

$$7f+x=14$$

The only value which satisfies the above equation is f=1,x=7

Erdos number at the end of fifth day,f+1,b,f+1,d,f+1,f,g,h On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	f+1	f+1
B	b	b	b
C	c	f+1	f+1
D	d	d	d
E	e	e = f+5	f+1
F	f(min)	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Erdos no. of C at the end is f+1 = 1+1 = 2. Hence option B.

34. C

Let us consider the Erdos number of A,B,C,D,E,F,G,H be a,b,c,d,e,f,g,h where f is the min, a is infinity.

At the end of 3rd day, F co authored with A and C. Since F has min Erdos number, the values of c,a will change to $f+1$ and the Erdos number of F will remain the same. [Because according to Erdos principle if a person co-authors with some one who has higher Erdos number then the Erdos number of co-authors will be min Erdos value + 1]

Average of the mathematicians is 3 Sum of the Erdos number of eight mathematicians=24

Erdos number at the third day: $f+1,b,f+1,d,e,f,g,h$

At the end of the fifth day, F co-authors with E thereby changing the average to 2.5 and the Erdos number of rest of the mathematicians remain unchanged. Sum of the Erdos numbers of eight mathematicians=20 So here the difference of $4[24-20]$ arose, which means e will be $f+5$ initially and changed to $f+1$ after co-authoring with F.

So the Erdos number at the third day: $f+1,b,f+1,d,f+5,f,g,h$

At the end of the third day, five mathematicians had the same Erdos number and the rest had distinct Erdos number from each other.

It cannot be $f+5$ because then there will be two mathematicians with the same Erdos number $f+1$.

So five mathematicians will have $f+1$, one with $f+5$, one with f , one with some different value say x

$$5(f+1)+f+5+f+x=24$$

$$7f+x=14$$

The only value which satisfies the above equation is $f=1,x=7$

Erdos number at the end of fifth day, $f+1,b,f+1,d,f+1,f,g,h$ On tabulating, we get

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	$f+1$	$f+1$
B	b	b	b
C	c	$f+1$	$f+1$
D	d	d	d
E	e	$e = f+5$	$f+1$
F	$f(\text{min})$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Hence erdos no. of E at the beginning of conference would be $f+5 = 6$.

35. B

Mathematician	At Start	End of 3rd day	End of 5th day
A	Infinity	$f+1$	$f+1$
B	b	b	b
C	c	$f+1$	$f+1$
D	d	d	d
E	e	$e = f+5$	$f+1$
F	$f(\text{min})$	f	f
G	g	g	g
H	h	h	h
		Total : 24 (as average = 3)	Total = 20 (as average = 2.5)

Since at the end of the 3rd day 5 people had identical erdos no. ($f+1$) so : $5*(f+1) + f+f+5+x = 24$; Only $f=1$ and $x = 7$ satisfies the equation. So out of 5 people who had identical erdos no. at the end of day 3, 2 of them had different nos. at the beginning. So there were $5-2 = 3$ participants who had the same Erdos number at the beginning of the conference.



Explanation [36 - 40]:

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

36. C

The question asks about the case where Chetan has sold 3 times and Michael sells only once.

Starting with Michael, for exactly one sell, the price should touch 120 only once as Michael sells the share only at price greater than 110.

if the price touches 120 twice or more, Michael will sell the share more than once which is not a desirable case.

Also, Chetan has to sell thrice consecutively which is only possible if the share price is 90 at one instance and rises to 120 in straight 3 days.

This is only possible in case 2. Hence the price on 3rd day's end in case 2 is 110.

VIDEO SOLUTION

37. E

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total Chetan will always sell 10 shares in all 5 days combined.

As Chetan has sold 10 shares, he'll get $110 \times 10 = 1100$ cash because of it in every case. Also Chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash Chetan will always have after 5 days = $1100 + 200 = 1300$.

The question asks about the case where Michael ended up with Rs 100 less cash than Chetan at the end of day 5.

So we have to look for the case where Michael has 1200 cash which is only possible when Michael has -10 number of shares in the end of day 5 and also has made profit of Rs. 100.

This is possible in case 7 as Michael has sold the shares at rs 120 but did not sell those shares as price never went below 90.

In the end Michael will have -10 shares at a price 110 which is Rs. 10 less than the price he sold at.

So he makes profit of 100 rs through it and will have the cash of 1100 through the sold shares.

Their difference in 7th case is $1300 - 1200 = 100$

Also, in this case they have same number of shares sold at the end of the 5th day. So E is the answer.

VIDEO SOLUTION

38. B

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total Chetan will always sell 10 shares in all 5 days combined.

As Chetan has sold 10 shares, he'll get $110 \times 10 = 1100$ cash because of it in every case. Also, Chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash Chetan will always have after 5 days = $1100 + 200 = 1300$.

The question asks about the case where Michael ended up with Rs 1300 cash less than Chetan.

As Chetan will have exactly 1300 cash in every case; we have to look for the cases where Michael does not make any profit and also does not have any sold shares at the end of 5 days.

This is possible in 4 cases ie. 3,4,5 and 9.

In each of these cases, the price fluctuates in the range 90-110 which does not allow Michael to buy or sell any shares.

Also, in all these cases the price of MCS is 100 at the end of day 4.

So the answer is 100.

VIDEO SOLUTION

39. D

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total Chetan will always sell 10 shares in all 5 days combined.

As Chetan has sold 10 shares, he'll get $110 \times 10 = 1100$ cash because of it in every case. Also, Chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash Chetan will always have after 5 days = $1100 + 200 = 1300$.

The question asks about the case where there has been the maximum possible increase in the combined cash balance of Chetan and Michael at the end of the fifth day.

As we know Chetan has 1300 cash in all the cases, so we have to maximize the case where Michael has the most cash.

Also, if we see clearly, the profit made by selling and buying is in hundreds while the cash received by selling the shares is much far in terms of cash.

As 10 shares sold give = $10 \times 110 = 1100$ cash. So we have to look at the case where Michael has sold most shares which is case 8.

In case 8 the price rises from 100 to 110 $\rightarrow$ 120 $\rightarrow$ 130 $\rightarrow$ 120 $\rightarrow$ 110

in this case 110(0 new shares) $\rightarrow$ 120(Michael sold 10 shares) $\rightarrow$ 130(Michael sold 10 shares) 120(Michael sold 10 shares) $\rightarrow$ 110(Michael does nothing) = 30 shares sold in total

Cash = 20 shares at 120 and 10 shares at 130 = $120 \times 20 + 130 \times 10 = 2400 + 1300 = 3700$

As Chetan has 1300 cash and Michael has 3700 cash;

total cash they have is : $3700 + 1300 = 5000$

VIDEO SOLUTION

40. A

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

To have 20 more shares than Chetan, Michael has to buy 10 shares which is case 1.

In case 1, the share price at the end of day 3 is 90.

VIDEO SOLUTION

100+ CAT Quant Questions



41. C

Let the toll charged at junctions A, B, C, and D be a, b, c and d respectively. Then the so that equal amount is collected through all route we have, $9+a+5=2+b+2+a+5=10+d+c=13+d$. Then from the options only option C satisfies the above equality. hence option C.

VIDEO SOLUTION

42. E

Let the toll charged at junctions A, B, C, and D be a, b, c and d respectively. Now since we want equal traffic through A and C, total cost through routes passing from A and C should be equal. So we have $(9+a+5) + (2+b+2+a+5) = (2+3+b+c+2) + (7+d+1+c+2)$. Only option E satisfy the above equality.

VIDEO SOLUTION

43. D

Now the fuel cost along different routes are :

SAT = 14

SBAT = 9

SBCT = 7

SDCT = 10

SDT = 13

Now, if we consider option D. Total cost for all routes comes out to be same which is 15. Hence option D.

VIDEO SOLUTION

44. A

Total cost = fuel cost + toll

Total cost along SAT : $14 + \text{tollA}$

Total cost along SBAT : $9 + \text{tollA} + \text{tollB}$

Total cost along SDT : $13 + \text{tollD}$

Now when option A is considered, total costs come out to be same.

Hence option A is correct.

VIDEO SOLUTION

45. C

The costs of the routes are as given below:

S - B - C - T = 7

S - B - A - T = 9

S - D - C - T = 10

S - D - T = 13

S - A - T = 14

Hence now 100% of the traffic flows through S - B - C - T

Now if we make the cost of traveling through S - B - C - T same as some other route not going through B, then the traffic will be equally distributed between these two routes. The lowest such route is S-D-C-T. The difference in cost = 3. Hence if we levy a toll of Rs.3 at B, the costs of SBCT and SBAT become 10,12 respectively and other routes are not affected. So 50% traffic flows through SBCT and 50% flows through SDCT. Hence cost in this policy = 10.

VIDEO SOLUTION

100+ CAT DILR Questions



46. E

A team which has K should have L also.

Since L is there in the team, N and U should not be there in the team. Since U is not there in the team, S and W should not be there in the team.

So, the team will have K, L, one of P and R and one of M or Q.

So, the team size will be 4.

VIDEO SOLUTION

47. E

Since N is in the team, L and K cannot be in the team.

The team can have one of M and Q. So, 2 ways of selection.

If the team has S, then it should have U and W as well.

If the team has no S, then it should have one of P or R.

So, the number of ways of forming the team is $2 \times (1+2) = 6$ ways

VIDEO SOLUTION

48. D

Out of P, R and S only 1 can be in the team. If S is there, U and W will also be there. So, P and R should not be in the team for its size to be maximum.

Out of M and Q, only 1 can be there.

If L is there in the team, N and U cannot be in the team.

If L is not there in the team, then K is also not there in the team but N and U can be in the team.

So, the maximum team size is 5 consisting of S, U, W, (M or Q), N.

[VIDEO SOLUTION](#)

49. C

Out of P, R and S only 1 can be in the team. If S is there, U and W will also be there. So, P and R should not be in the team for its size to be maximum.

Out of M and Q, only 1 can be there.

If L is there in the team, N and U cannot be in the team.

If L is not there in the team, then K is also not there in the team but N and U can be in the team.

So, the maximum team size is 5 consisting of S, U, W, (M or Q), N.

So, M can be a member of team size 5.

[VIDEO SOLUTION](#)

50. A

If L is in the team, the team should include K also. The team should have one among P, R and S and one among M and Q.

So, the team size cannot be 3 if L is in the team.

[VIDEO SOLUTION](#)

100+ CAT VARC Questions



Verbal

51. C

From the 3rd para we find the sentences like '... Mr. Lindblad and the Council of Europe adopt as fact the wildest estimates of those "killed by communist regimes" (mostly in famines) from..'. From here we can infer that among all apprehensions that Mr. Goran Lindblad expresses against communism this option C is admitted indirectly.

[VIDEO SOLUTION](#)

52. B

Options A, C and E are irrelevant according to the passage.

Both B and D seem to be the answers, but the question asks for the real reason.

If communism is not a threat, then it is not required to destroy it. So, D cannot be the real reason.

Hence, option B is the answer.

[VIDEO SOLUTION](#)

53. E

Option A, B, C are clearly not the answers and are out of context. Out of D and E, option E gives perfect reason of why author cites examples of atrocities perpetrated by European colonial regimes.

[VIDEO SOLUTION](#)

54. A

{This was a system of racist despotism, which dominated the globe in Stalin's time. And while there is precious little connection between the ideas of fascism and communism, there is an intimate link between colonialism and Nazism. The terms lebensraum and konzentrationslager were both first used by the German colonial regime in South-West Africa (now Namibia), which committed genocide against the Herero and Nama peoples and bequeathed its ideas and personnel directly to the Nazi party.}

The term '*racist despotism*' coupled with the information on genocide adequately conveys the tyranny of one race/group over another. The author attempts to compare the potential motives behind the atrocities and identifies that the ideas of colonialism and Nazism are similar in that they constitute examples of tyranny of one race over another. Option A correctly captures this.

Option B is incorrect since we do not have enough information on the scale of events: we cannot definitively compare the magnitude of the genocides. Options C, D and E are neither implied nor discussed in the passage.

Hence, the correct choice is Option A.

[VIDEO SOLUTION](#)

55. D

Except D, all the other options are compelling reasons for the silence of Council of Europe over colonial atrocities.

Option D is the compelling reason for condemnation of communism.

Hence, option D is the answer.

[VIDEO SOLUTION](#)

CAT Expected Questions PDF



56. C

A just society is not a utopia according to the passage => option A is wrong.

The society mentioned in the passage is not hypothetical => options B and E are wrong.

Between option C and option D, option C fits better to be the answer because it is similar to the authors views in the passage whereas option D is too generalized.

Hence, C is the answer.

[VIDEO SOLUTION](#)

57. A

It is clearly mentioned in the 2nd para of the passage that the original agreement is a hypothetical situation and is not influenced by position or status of the individual. Refer to the sentence "It is understood as a purely hypothetical situation.....intelligence, strength and the like."

[VIDEO SOLUTION](#)

58. D

Consider the lines "Among the essential features of this situation is that no one knows his place in society, his class position or social status, nor does anyone know his fortune in the distribution of natural assets and abilities, his intelligence, strength, and the like. I shall even assume that the parties do not know their conceptions of the good or their special psychological propensities. The principles of justice are chosen behind a veil of ignorance". Thus, the author feels that for the rules to be truly unbiased, the rule-makers should have no information about their natural advantages (i.e. social position, strengths etc). Thus, we have to select an option where the rule makers have no idea about their own individual advantages.

Of the given options, only option D presents such a situation. The rule-makers are not aware of where they will be born or what their capabilities will be. Hence, option D.

In the case of the businessmen, irrespective of whether they can return to society, they are aware of their individual strengths. Same is the case with the immigrants. Even if they don't know what resources are needed, they know their own capabilities. In the case of the students, we don't know if they are unaware of where they are born and their natural advantages. Hence, the right option is option D.

[VIDEO SOLUTION](#)

59. B

Refer to the sentence "Our social situation is just.....general system of rules which define it." Option B justifies this whereas other options deviate from this statement.

[VIDEO SOLUTION](#)

60. D

The author says that the initial equality and the veil of ignorance are the most important.

Among the given options, only option D fits to be the answer because of the initial equality it poses.

Hence, option D is the answer.

[VIDEO SOLUTION](#)



CAT Important Topics



61. B

It has been stated in the passage that "For the critical attitude is not so much opposed to the dogmatic attitude as super-imposed upon it: criticism must be directed against existing and influential beliefs in need of critical revision - in other words, dogmatic beliefs. A critical attitude needs for its raw material, as it were, theories or beliefs which are held more or less dogmatically."

From this we can infer that critical attitude is not opposed to dogmatic beliefs. Rather it uses dogmatic beliefs as raw material. Hence, we can eliminate options A, D and E.

From the passage, we can infer that the critical attitude acts on dogmatic beliefs to refine them. Hence, option B is more appropriate.

[VIDEO SOLUTION](#)

62. A

Options C,D,E do not hold true.

Amongst option A and B, option A is more appropriate. Refer to the 3rd paragraph. It is written that

"the pseudo-scientific attitude is more primitive than, and prior to, the scientific attitude: that it is a pre-scientific attitude. And this primitivity or priority also has its logical aspect. For the critical attitude is not so much opposed to the dogmatic attitude as super-imposed upon it: criticism must be directed against existing and influential beliefs in need of critical revision — in other words, dogmatic beliefs. A critical attitude needs for its raw material, as it were, theories or beliefs which are held more or less dogmatically".

63. D

Refer to the lines:

It further suggests that genetically speaking the pseudo-scientific attitude is more primitive than, and prior to, the scientific attitude: that it is a pre-scientific attitude. And this primitivity or priority also has its logical aspect. For the critical attitude is not so much opposed to the dogmatic attitude as super-imposed upon it.

It clearly illustrates option 4.

64. E

Refer to the last 3 lines

"From the point of view here developed, all laws, all theories, remain essentially tentative, or conjectural, or hypothetical, even when we feel unable to doubt them any longer. Before a theory has been refuted we can never know in what way it may have to be modified."

Option E clearly supports this argument.

Option A contains a distortion of "endless questioning". A critical attitude requires questioning but not endless questioning.

Option B has not been implied anywhere in the passage.

The first part of option C is true but the second part is false. The result of a critical attitude is not noise but tested beliefs that are subject to change.

Option D has not been implied anywhere in the passage.

65. C

Refer to these lines

"But the distinction between dogmatic and critical thinking, or the dogmatic and the critical attitude, brings us right back to our central problem. For the dogmatic attitude is clearly related to the tendency to verify our laws and schemata by seeking to apply them and to confirm them, even to the point of neglecting refutations, whereas the critical attitude is one of readiness to change them — to test them; to refute them; to falsify them, if possible. This suggests that we may identify the critical attitude with the scientific attitude, and the dogmatic attitude with the one which we have described as pseudo-scientific."

Option C is the main point of the above paragraph. Hence, option C is the answer.



66. D

The first statement says "It can feel like a treadmill that gets you nowhere". This is an opinion and hence the statement is a judgement.

Statement 2 is also a personal opinion and hence a judgement.

Similarly, statements 3 and 4 are also opinions and hence can be categorised as judgements.

67. A

The first statement is an opinion since it says "Given the poor quality..." and "...should be switching". The second statement is clearly a fact. The fourth statement is a judgement since the words "...ironic it is..." indicate the opinion of the author and not something that can be verified by facts. The answer is option A.

68. C

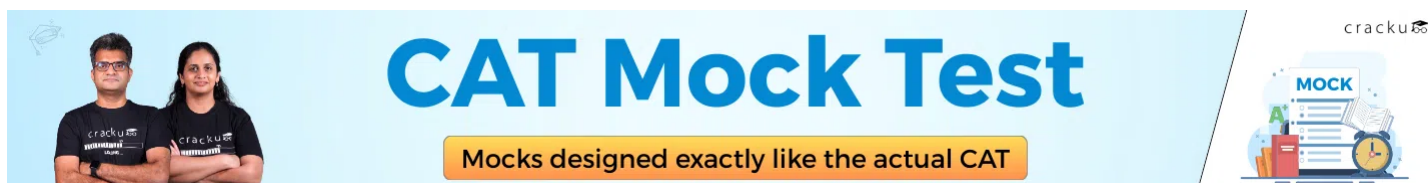
Statement 1 is an inference, because it is drawing a conclusion (Sarva Shiksha Abhiyan has managed to keep pace with its ambitious goals) based on a fact (all statistical indications). Statement 2 is a judgement since "the vital link between healthy bodies and healthy minds" is not something that can be investigated for data. Statement 3 is clearly a fact. Statement 4 is an opinion, and hence it is a judgement.

69. E

Statement 1 expresses the personal opinion of the author and hence it is a judgement. Statement 2 is a fact. Statement 3 is again an opinion of the author and hence is a judgement. Statement 4 is an inference because it draws a conclusion ("has the potential...") based on a fact ("Every red tape procedure is a point of contact with an official..."). Option E is the answer.

70. B

Statements 1 and 3 are judgements because they are both opinions of the author. This eliminates all options except B. Hence B is the right answer.



71. C

The last sentence should talk about the statement that the author made to the art director. According to the author, the statement is not a rule. So, option a) and c) are applicable. Between these two, option c) is better because it specifically talks about this statement and not about "rules" in general. So, c) is the best concluding sentence.

72. E

Option a) talks only about one part of the chain comprising of factory, dealer and customer. Option b) is far-fetched. Options c) and d) are out of scope. Option e) completes the para in the best possible way.

73. B

The last sentence in the paragraph should counter the statement made by the nations as well as talk about diplomacy, to bring the para to a meaningful conclusion. This is best captured in option b), which says, when the veil is lifted, diplomacy is often used as a disguise for the rule of force.

74. **D**

The para talks about the relationship between age and entrepreneurial spirit. When people are young, they tend to believe that age brings in more credibility in transmitting information to others. However, as people age, they become less receptive to information and their opportunity costs rise. Hence, it is very likely that they become reluctant to exploit entrepreneurial opportunities as they age. Statement D, therefore, is the best concluding statement.

75. **A**

The para talks about good maps throwing away a large amount of non-essential information and retaining more important information. They are also simplified as a result. So, the last sentence should be about good theories being simplified and more valuable. Option a) captures the essence in the best way.

A promotional banner for Cracku CAT Online Coaching. On the left, a man and a woman wearing Cracku t-shirts stand with their arms crossed. In the center, the text "CAT Online Coaching" is written in large, bold, black letters. Below this, a red button contains the white text "Your road to IIM starts here". On the right, there is an illustration of a person wearing a headset and working on a laptop, with the Cracku logo above it.

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